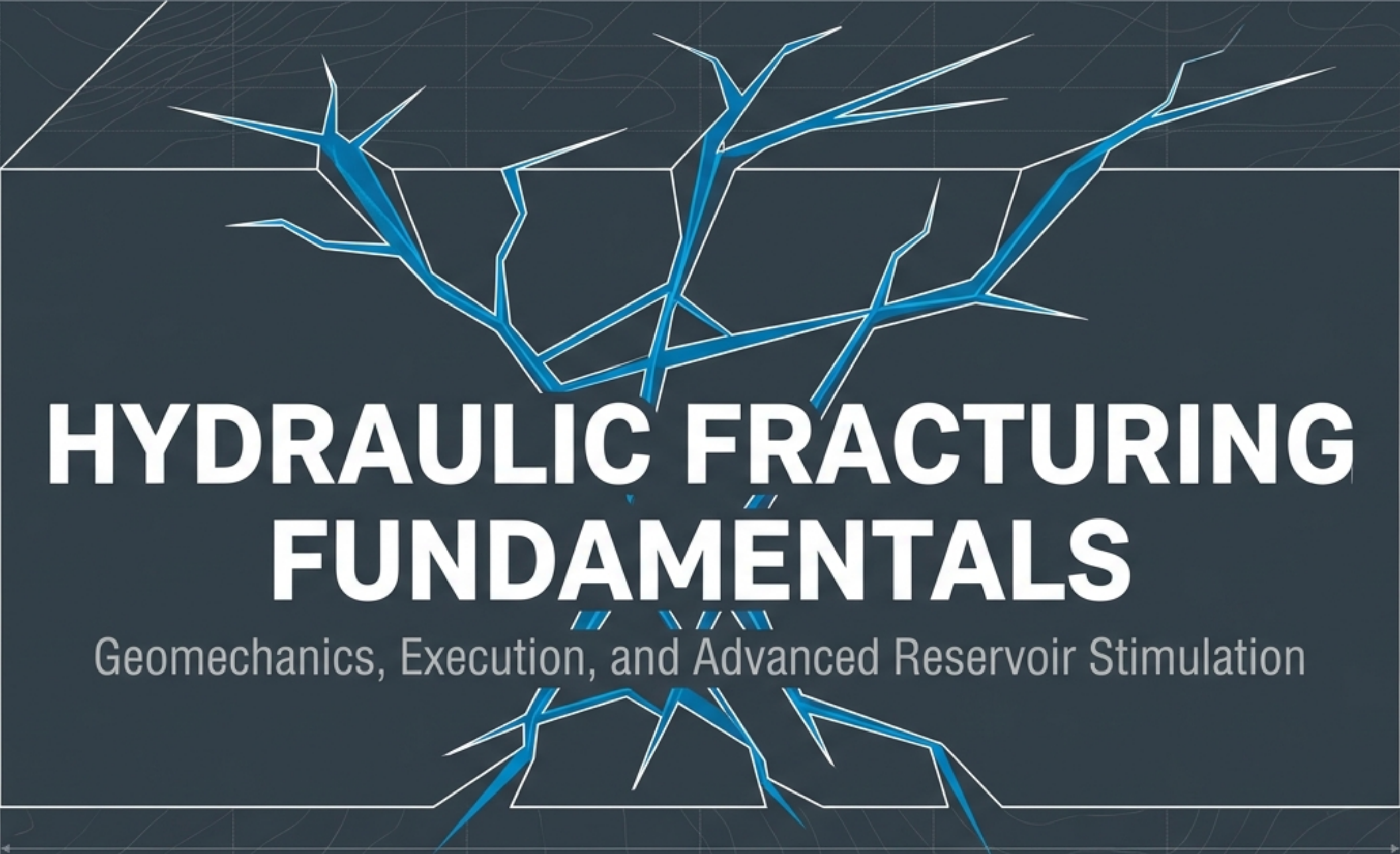




HYDRAULIC FRACTURING FUNDAMENTALS

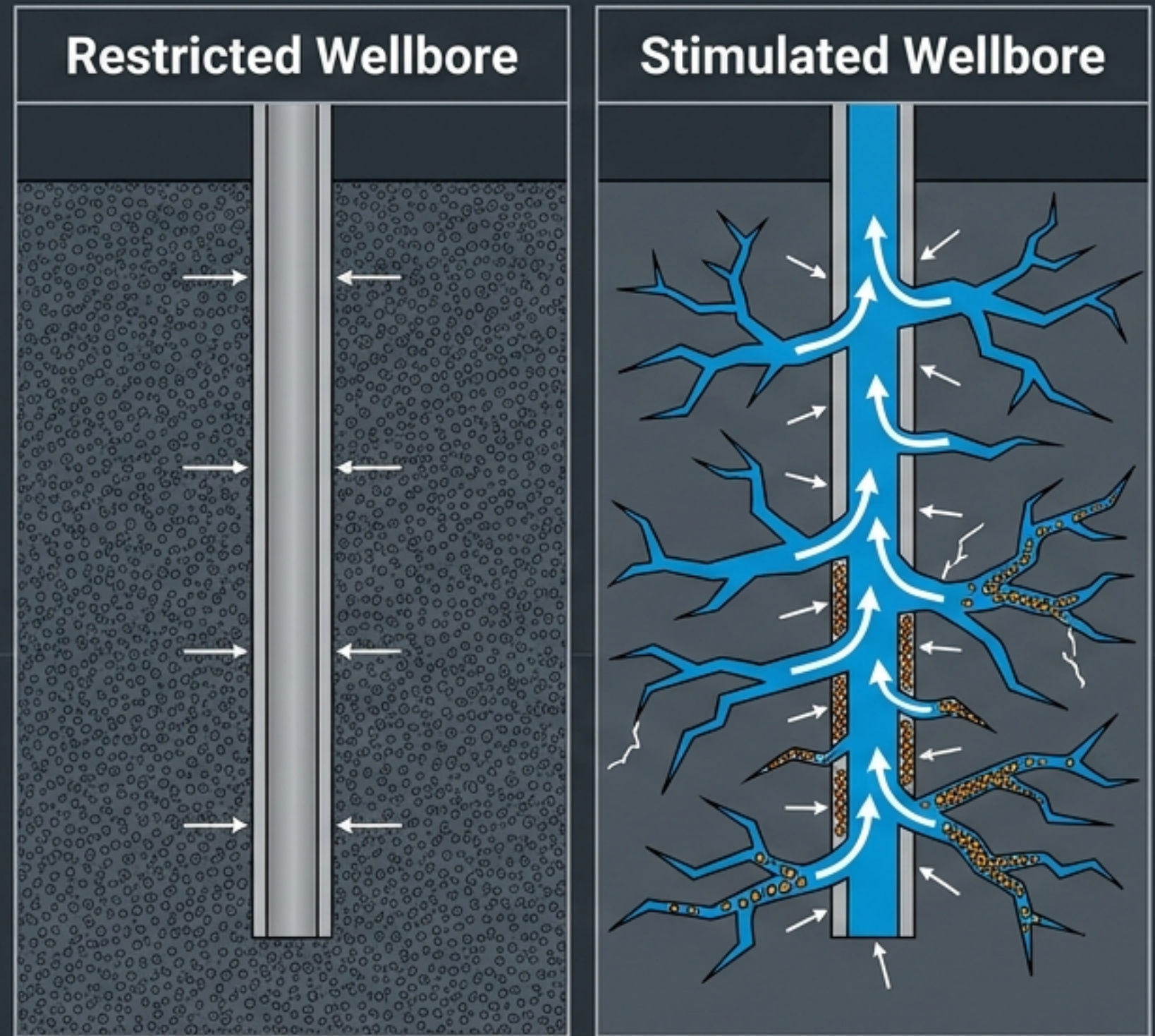
Geomechanics, Execution, and Advanced Reservoir Stimulation

Presented by Engineer Sajjad k.hano

The Core Concept

Definition

Hydraulic Fracturing is a well stimulation technique designed to maximize productivity in low-permeability (tight) reservoirs or to bypass near-wellbore formation damage.



Mechanism: Mechanically failing the rock matrix via high-pressure specialized fluids to create highly conductive pathways.

The In-Situ Stress Regimen

Fracture initiation and orientation are not random; they are strictly governed by localized subsurface stresses.

Vertical Stress (σ_v)

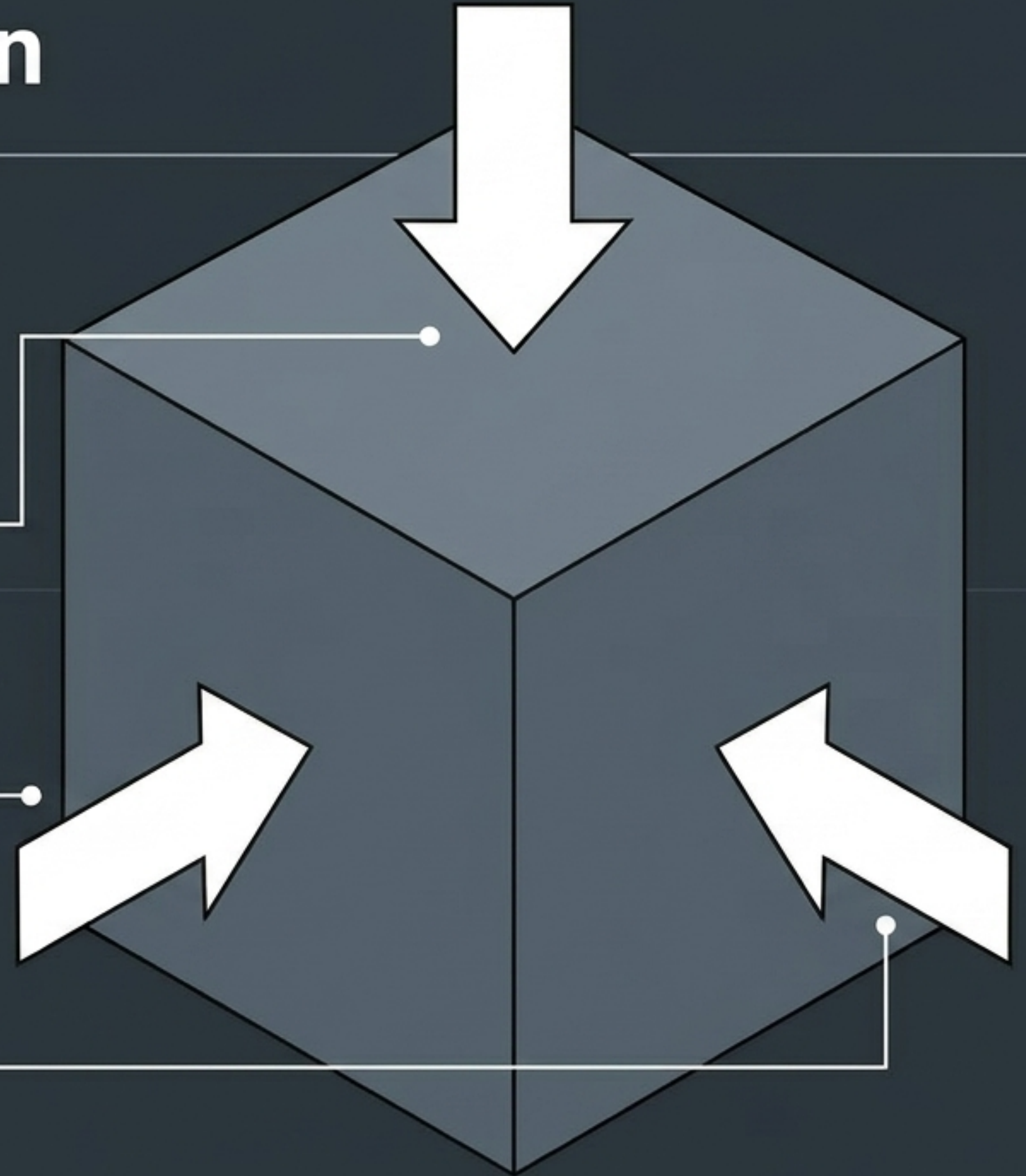
At typical reservoir depths, this downward vector is the dominant principal stress.

Maximum Horizontal Stress ($\sigma_{H_{max}}$)

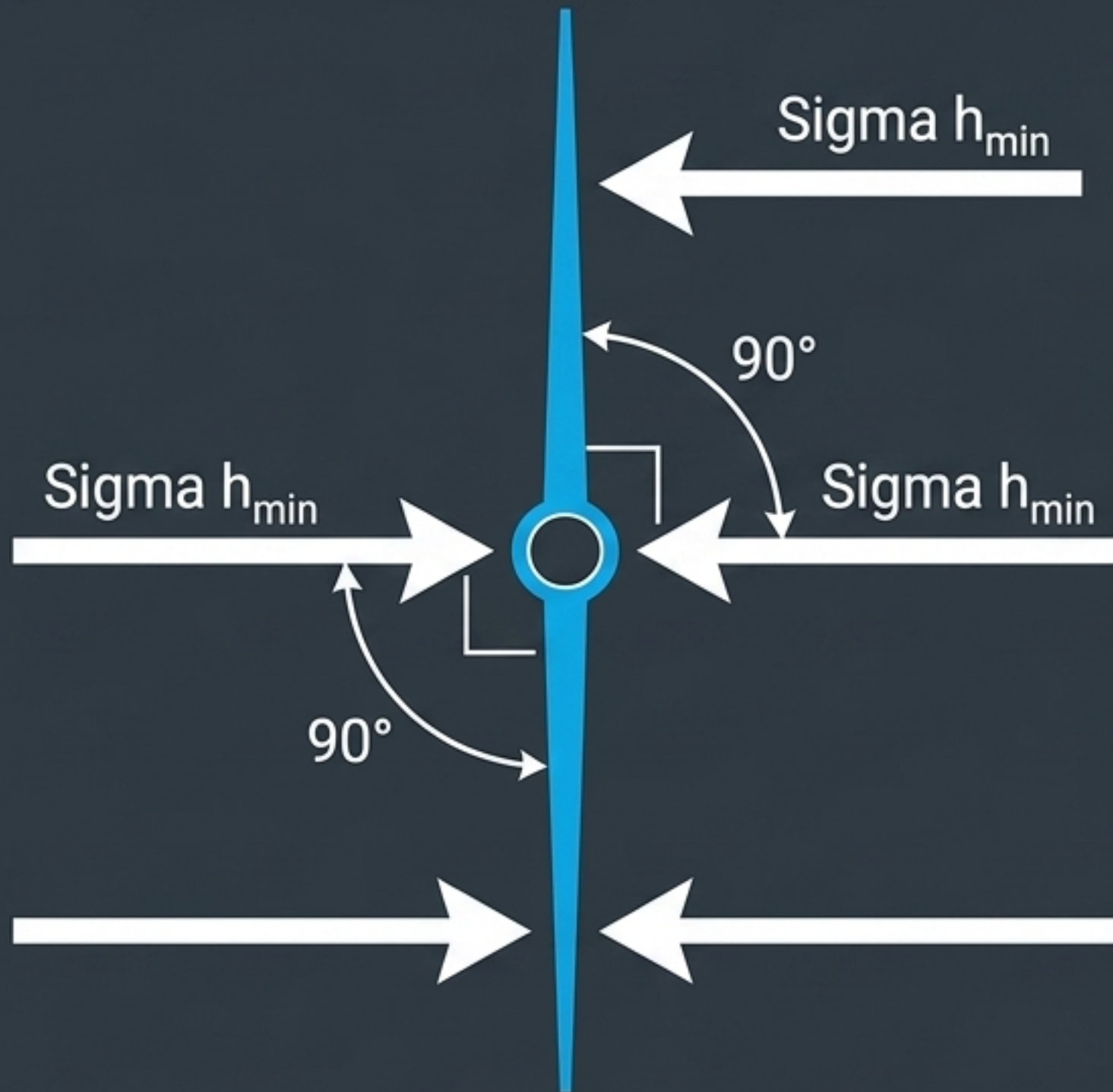
The secondary inward horizontal compressive force.

Minimum Horizontal Stress ($\sigma_{h_{min}}$)

The weakest compressive force. Fractures will naturally open against this axis.



Fracture Orientation: The Path of Least Resistance



Structural Mechanics

Fractures naturally propagate along the path of least resistance. They will always open strictly perpendicular to the minimum horizontal stress ($\text{Sigma } h_{\min}$).

Propagation Rule

Because vertical stress ($\text{Sigma } v$) is the principal, strongest stress at typical reservoir depths, fractures are forced to open horizontally and propagate vertically and longitudinally through the formation.

Breakdown Pressure Mechanics

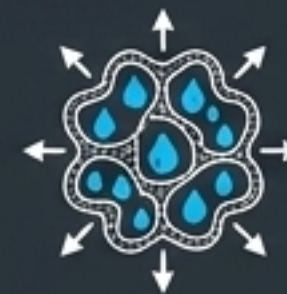
The critical pressure threshold required to initiate a fracture under elastic rock conditions. The injected fluid must overpower both the earth's stress and the rock's structural integrity.

Breakdown Pressure (P_b)



T_o (Tensile Strength of the Rock)

The natural cohesive strength binding the rock matrix together. Illustrated as an interlocking microscopic rock structure resisting separation.

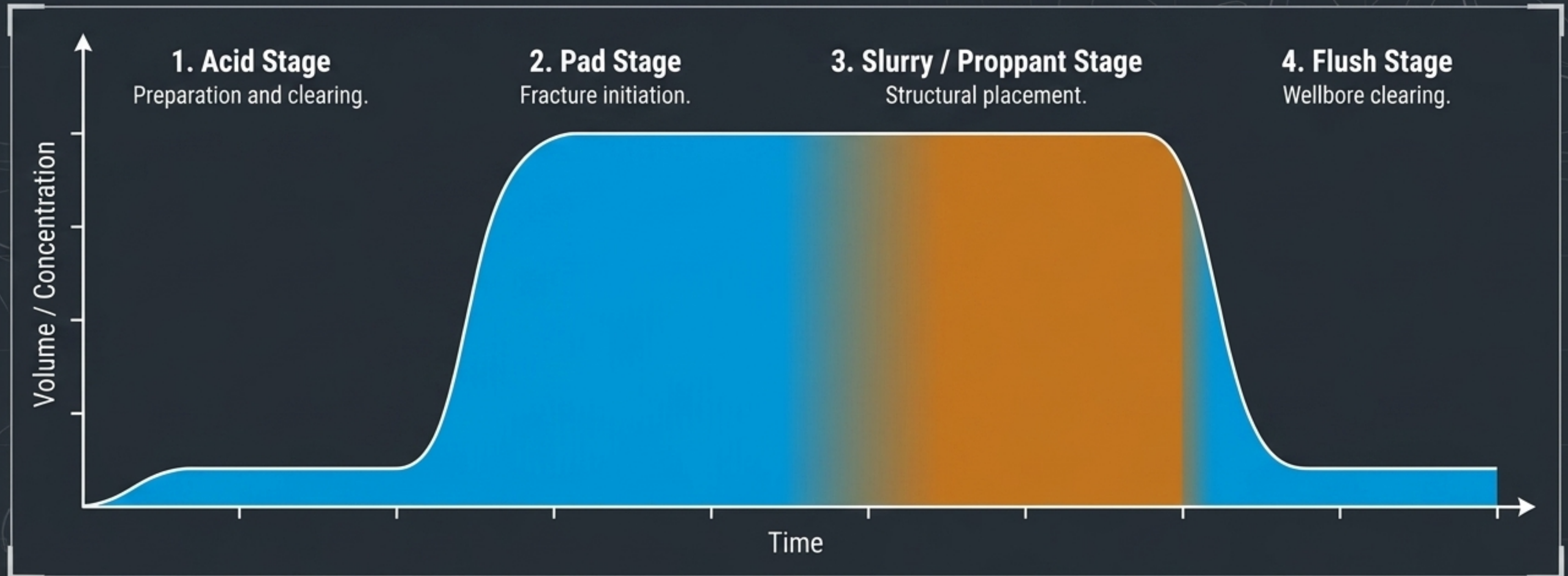


P_p (Pore Pressure)

The internal pressure of the fluids already residing within the rock's pore spaces.

The Pumping Sequence: Operational Overview

A typical hydraulic fracturing operation is executed not as isolated events, but as continuous, perfectly sequenced fluid stages.



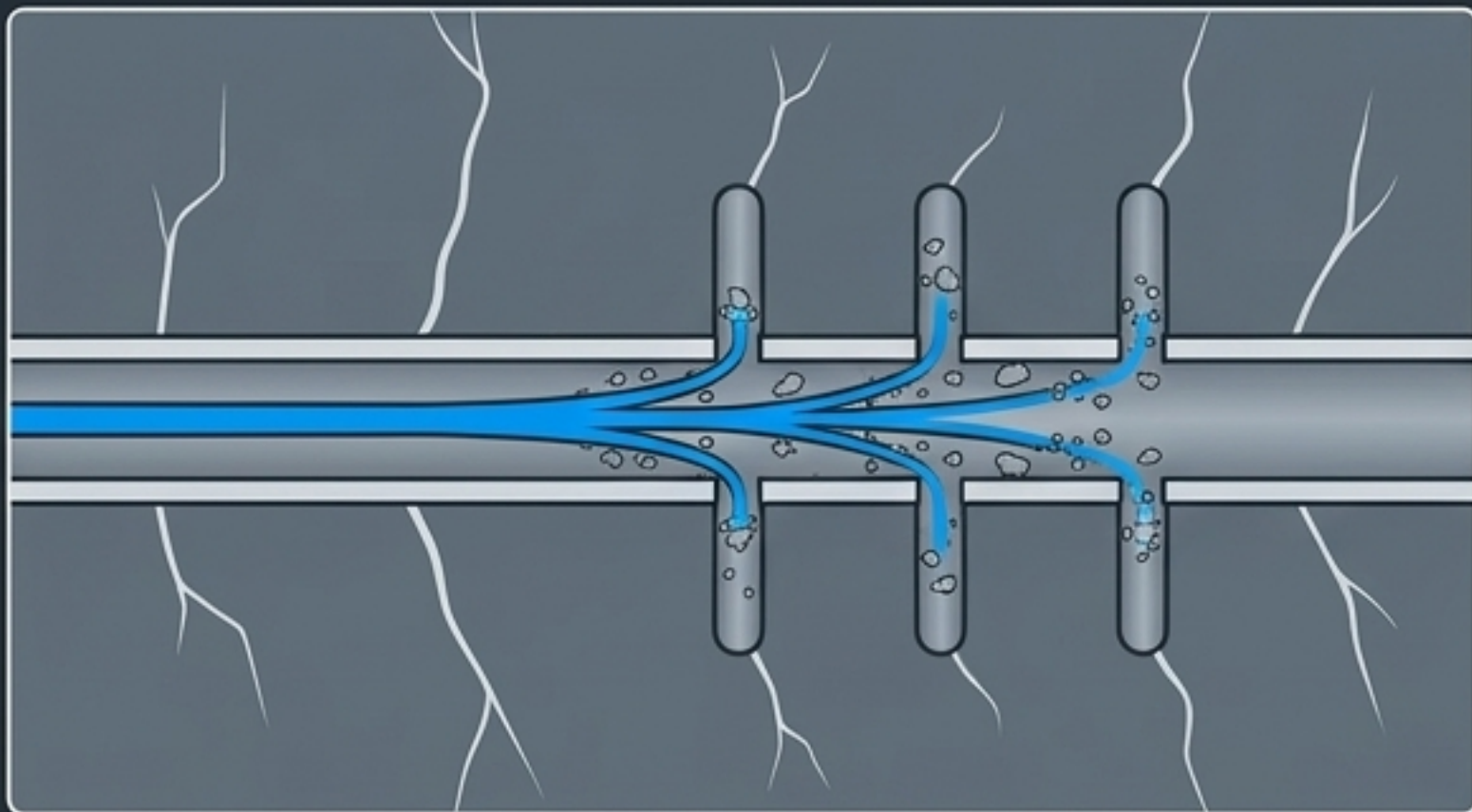
Execution Phase I: Initiation & Propagation

Acid Stage

Component: 15% HCl (Hydrochloric Acid).

Mechanism: A small volume is pumped directly into the perforations.

Purpose: Clears perforation debris and dissolves scale, directly reducing the breakdown pressure required for the next step.

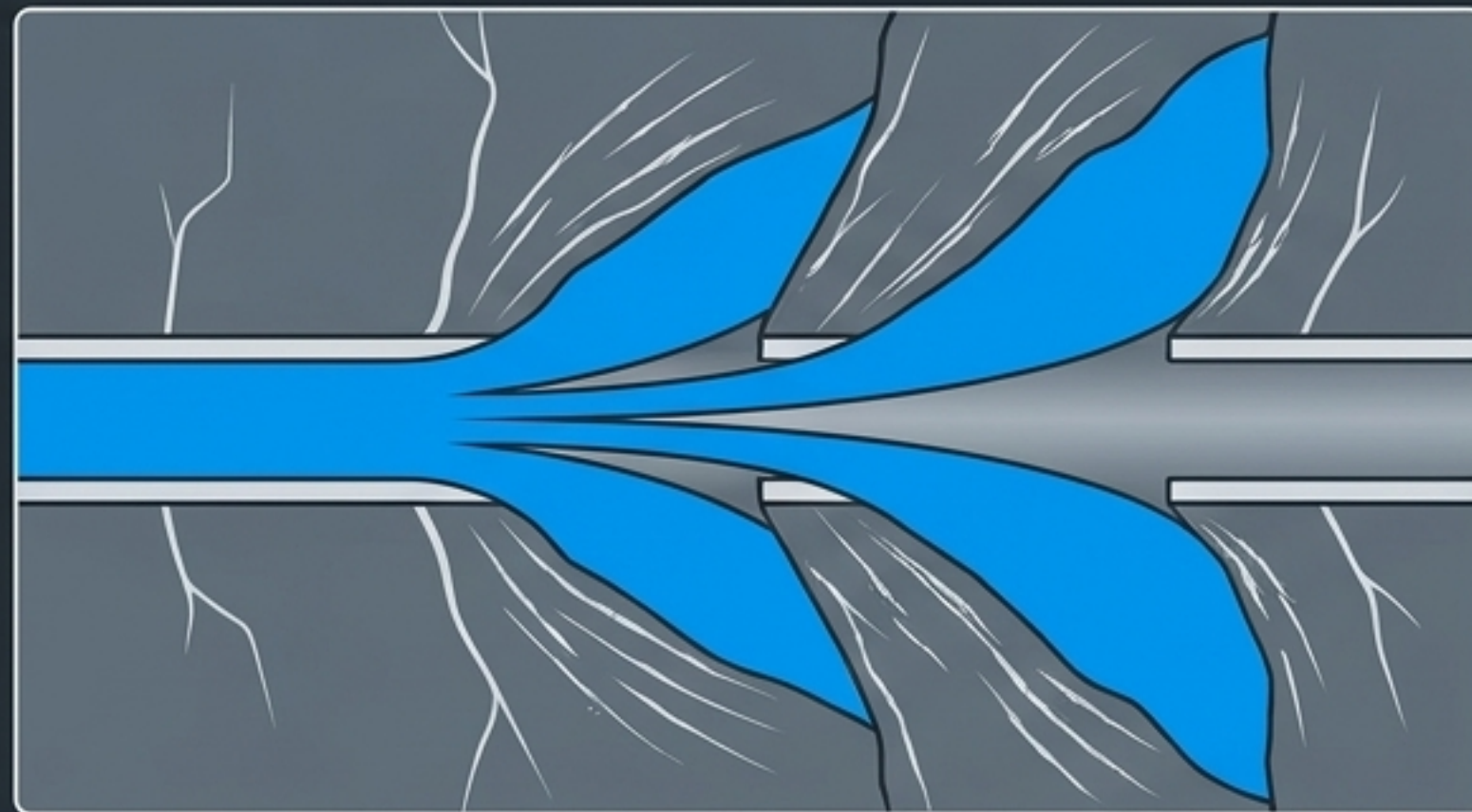


Pad Stage

Component: High-viscosity fluid, strictly 0 proppant.

Mechanism: Injected at massive pressures exceeding Breakdown Pressure.

Purpose: Initiates the physical fracture and hydraulically builds the critical width and length required to accept proppant.



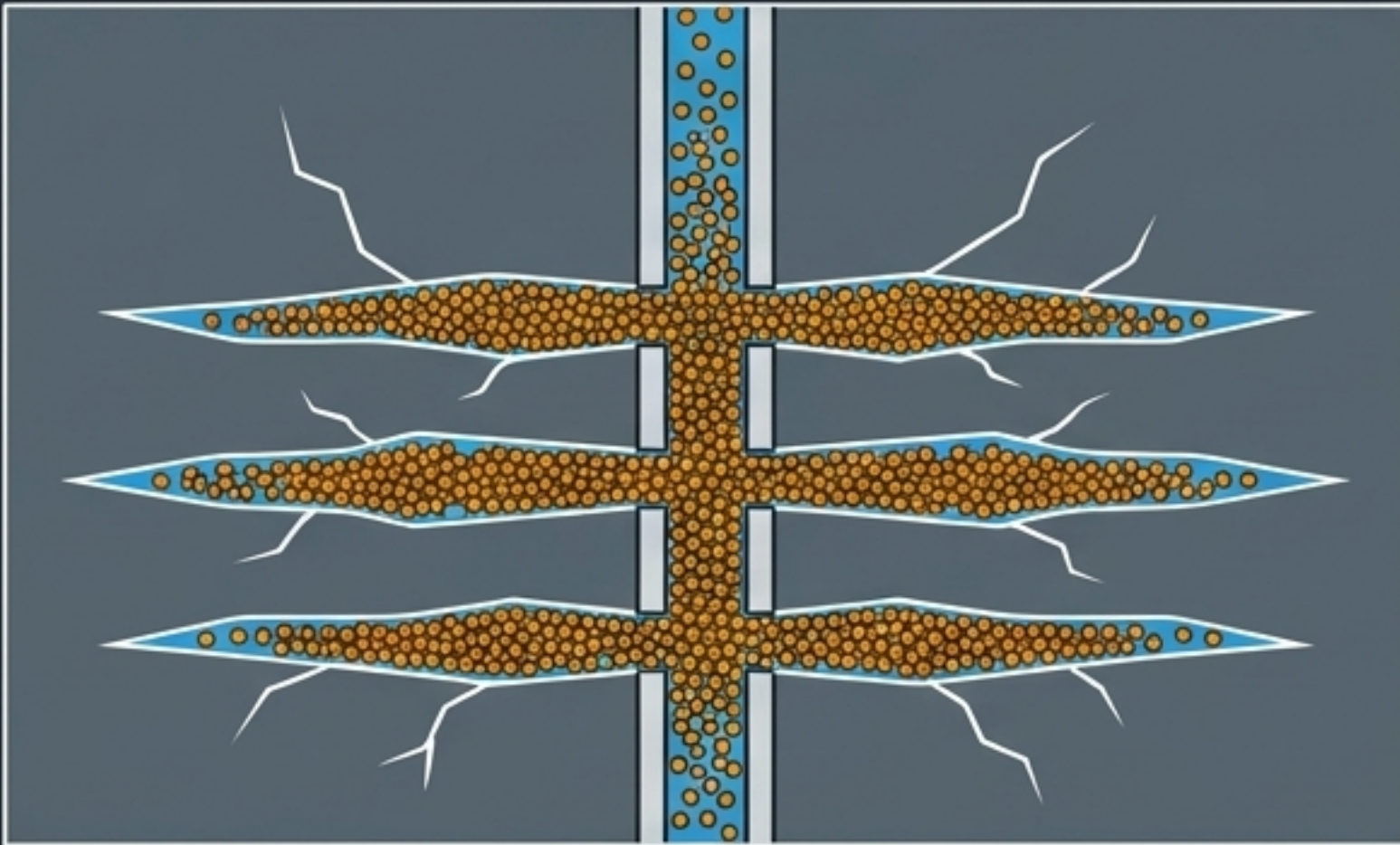
Execution Phase II: Placement & Cleanup

Slurry / Proppant Stage

Component: Sand or ceramic beads suspended in fluid.

Mechanism: Introduced in stepped, ramping increments (e.g., from 0.5 up to 6+ ppa).

Purpose: Places the necessary solid material deep into the fracture network to keep it unclosed once pumping ceases.

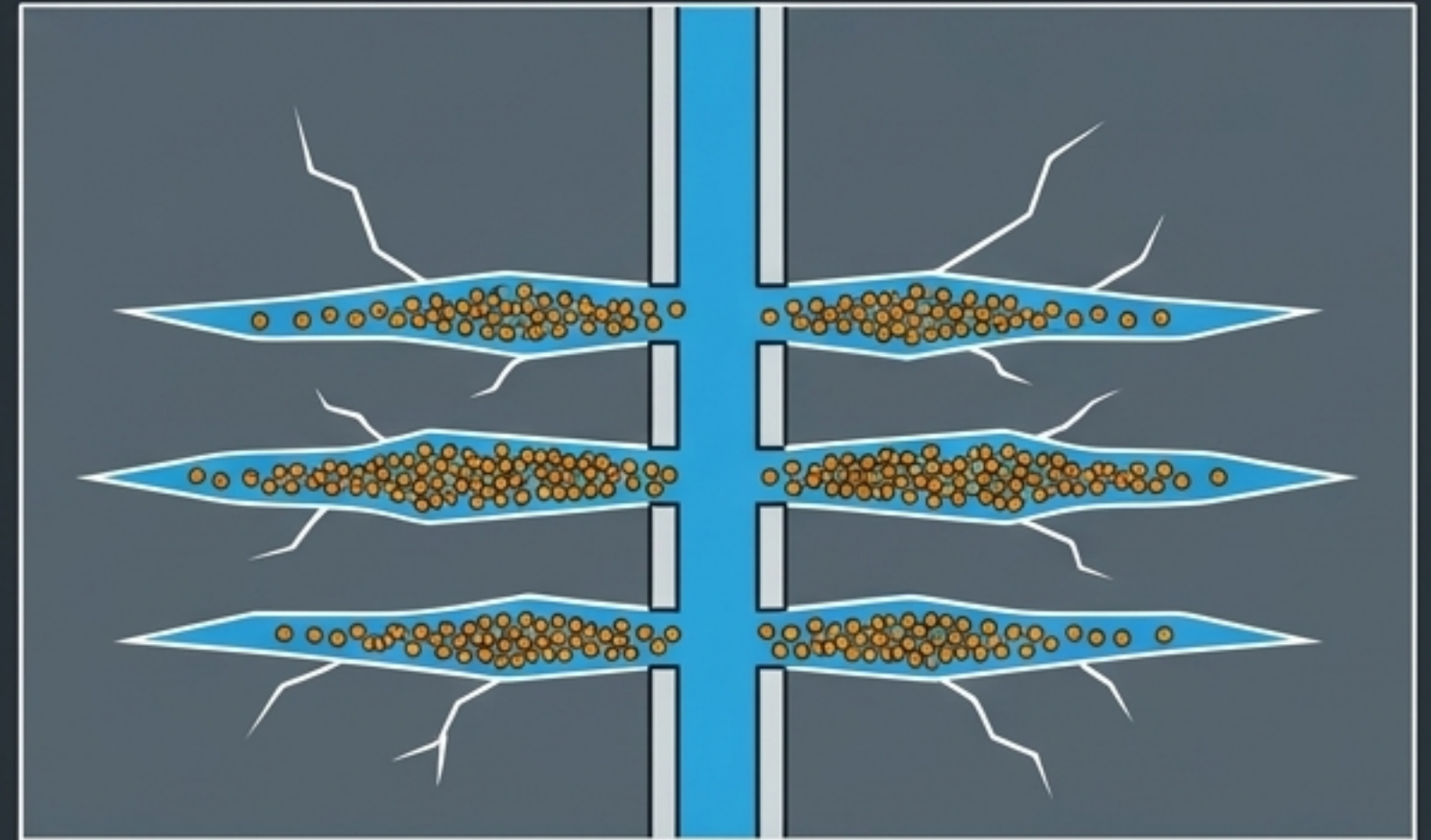


Flush Stage

Component: Fresh fluid.

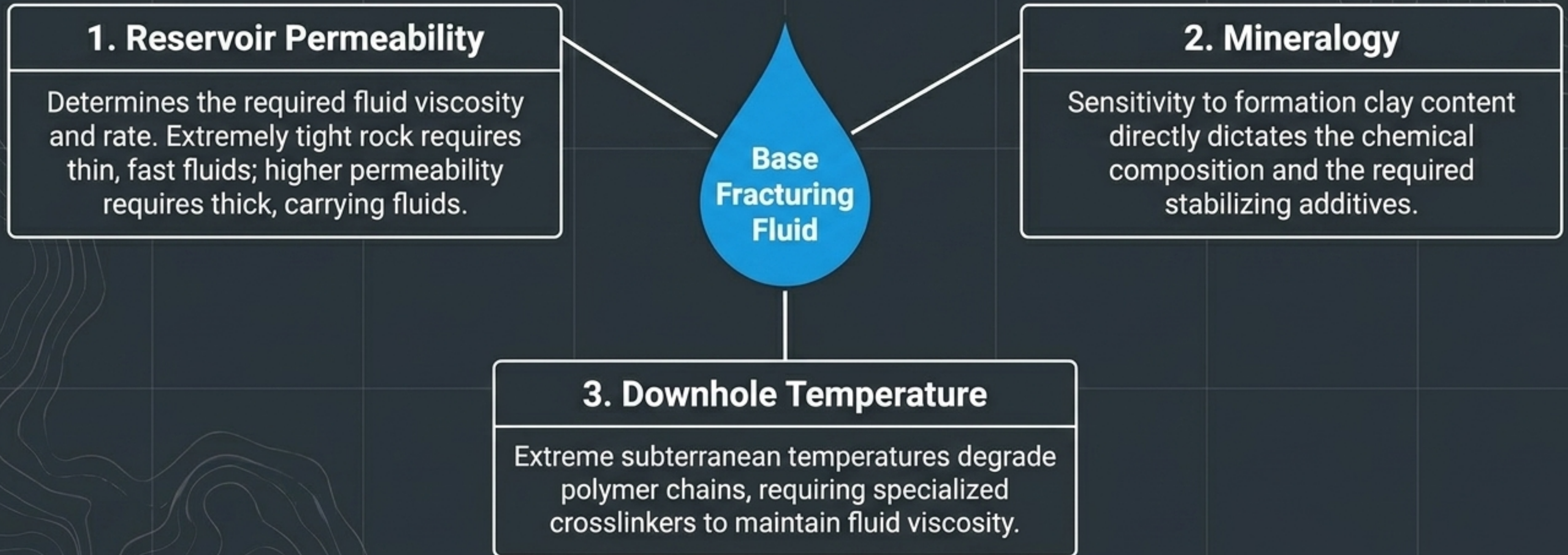
Mechanism: Pumped exactly to the volume of the wellbore casing.

Purpose: Pushes the remaining slurry entirely out of the wellbore and into the formation, ensuring the pipe remains perfectly clear for production.



Fracturing Fluid Systems: Engineering the Carrier

The carrier fluid is the vehicle for both hydraulic energy and proppant. Selection is strictly dictated by three primary reservoir conditions.



Fluid Diagnostic Matrix I: Slickwater & Linear Gels

Slickwater

Main Components: Water + Friction Reducer + Biocide.

Key Characteristics: Low viscosity, pumped at extremely high rates. Creates highly complex, narrow fracture networks.

Common Applications: Tight gas and shale reservoirs characterized by extreme high brittleness.

Linear Gels

Main Components: Water + Guar or HEC polymers.

Key Characteristics: Moderate viscosity, providing improved proppant transport capabilities compared to slickwater systems.

Common Applications: Matrix cleanup and moderate-permeability sandstone formations.

Fluid Diagnostic Matrix II: Crosslinked & Energized Systems

Crosslinked Gels

Main Components: Linear Gel base + Borate or Zirconium crosslinkers.

Key Characteristics: Extremely high viscosity with excellent proppant carrying capacity. Engineered to create exceptionally wide fractures.

Common Applications: Deeper, higher-temperature reservoirs and high-permeability contrast zones.

Energized / Foam Systems

Main Components: CO₂ or N₂ gas + Surfactant.

Key Characteristics: Utilizes minimal liquid volume allowing for rapid cleanup and minimized formation damage.

Common Applications: Highly water-sensitive formations and depleted reservoirs with low formation pressure.

Proppant Mechanics: Defeating Closure Stress

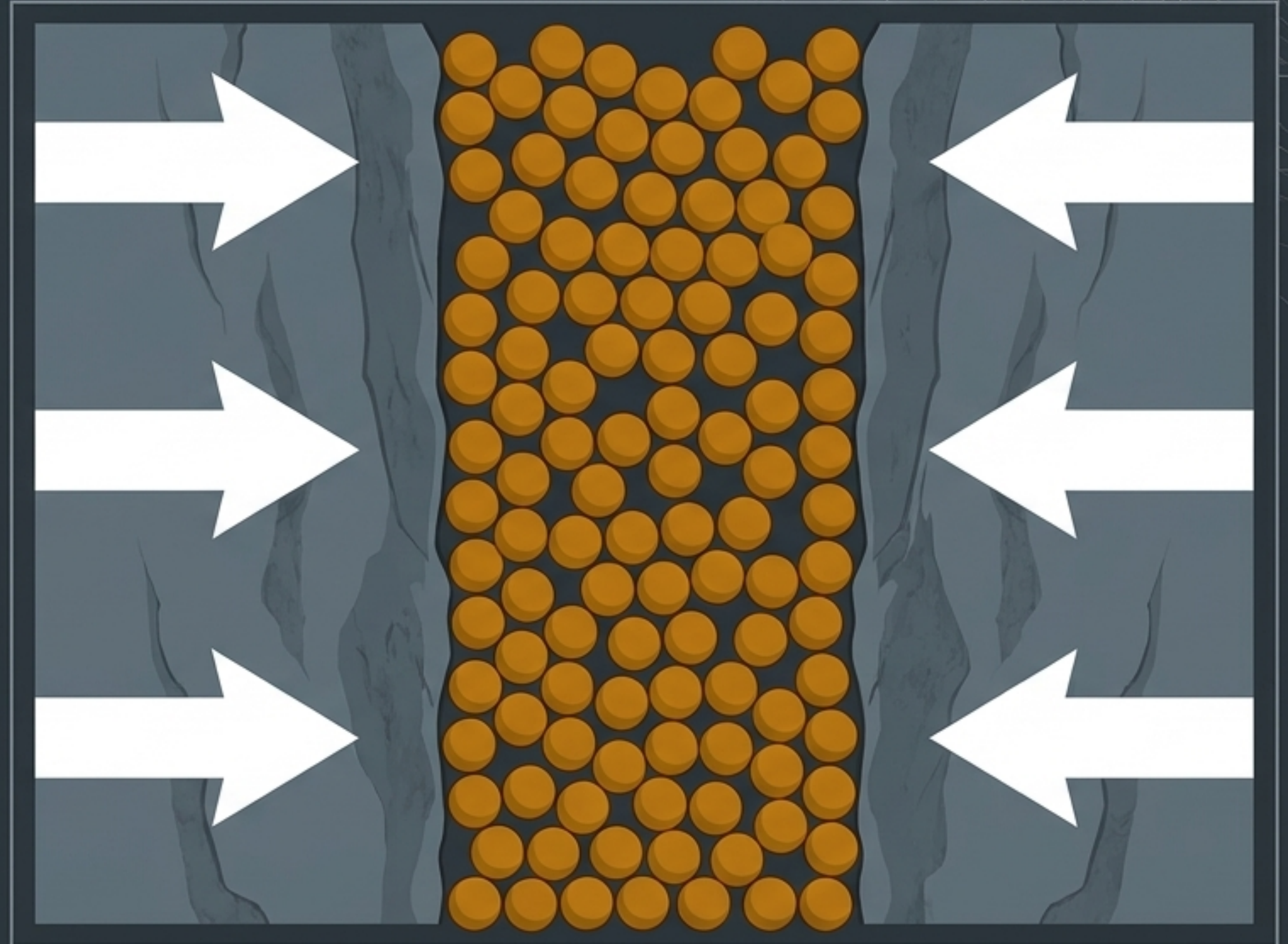
The Physical Reality

Once pumping stops and hydraulic fluid pressure bleeds off, the reservoir's immense closure stress aggressively attempts to snap the newly created fracture shut.

The Ultimate Objective: Fracture Conductivity

Proppants are placed to permanently prop the fracture open.

Fracture Conductivity =
(Fracture Width) x (Proppant Permeability)



Proppant Material Science

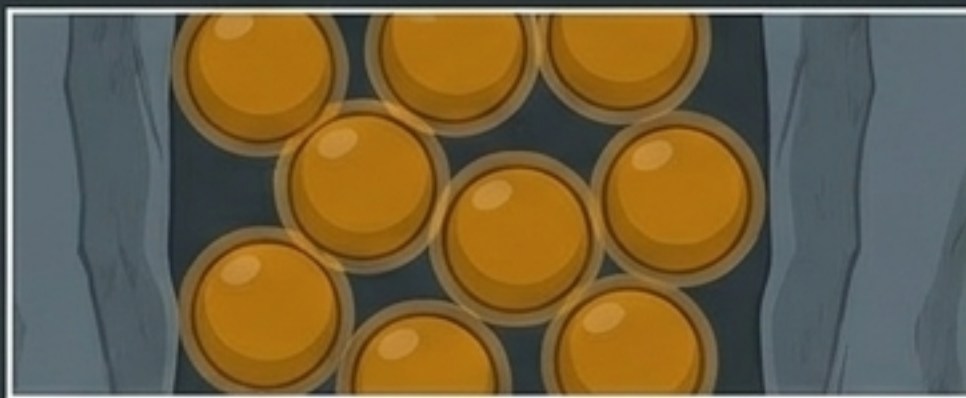
Unconsolidated Silica Sand

The baseline material. The most common and highly cost-effective solution for shallow reservoirs with standard closure stresses.



Resin-Coated Sand

Mid-tier engineered material. Provides significantly higher compressive strength. The resin coating allows grains to bond together, providing excellent flowback control.



Sintered Bauxites / Ceramics

Top-tier synthetic material. Engineered specifically for extreme subterranean environments, capable of withstanding immense closure stresses over 8,000 psi without crushing.



Low Closure Stress

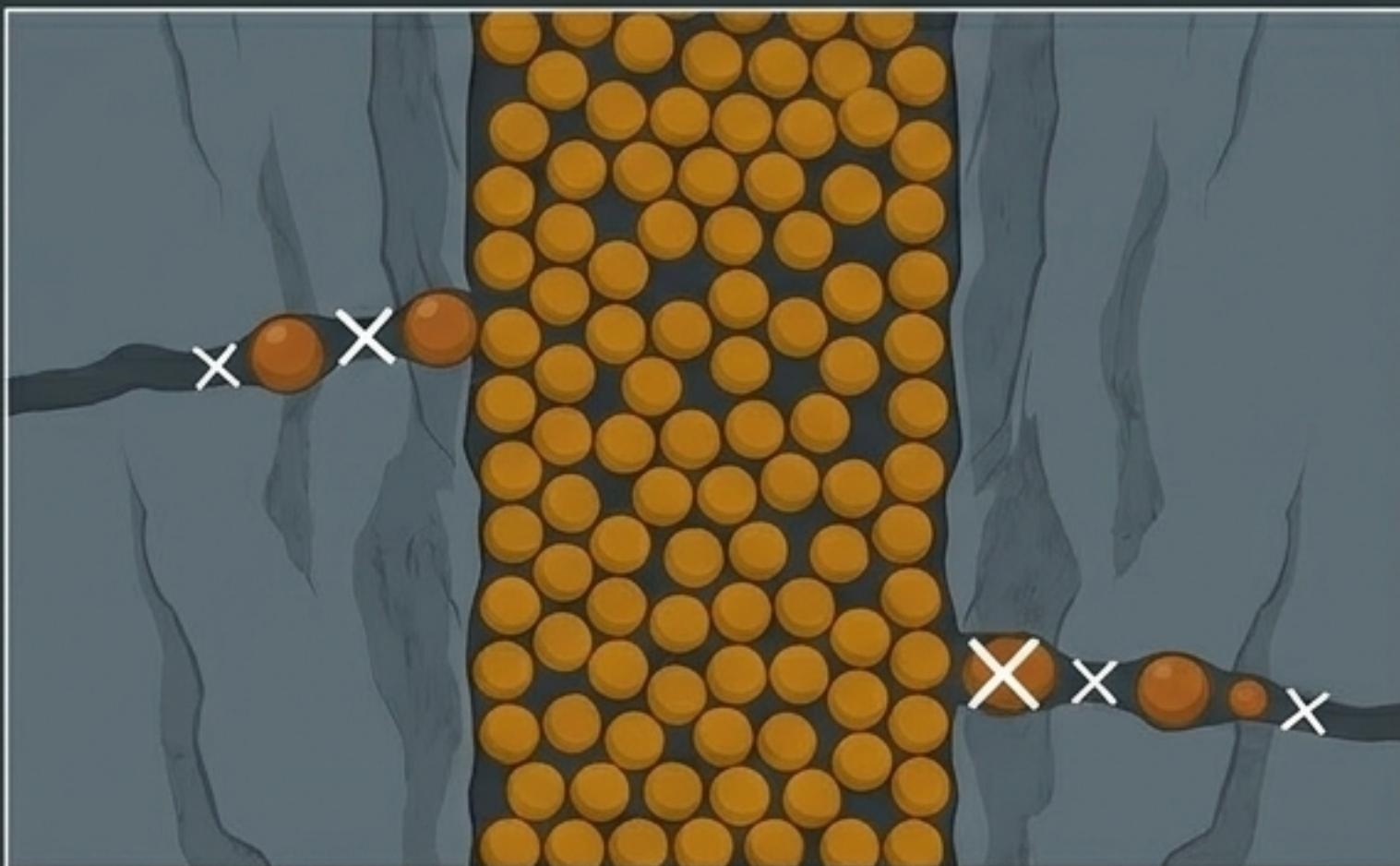
Extreme Closure Stress (> 8,000 psi)

Proppant Sizing & Mesh Dynamics

Proppant size is measured by Sieving/Mesh Size. Selection is a delicate balance between maximum flow capacity and maximum physical penetration.

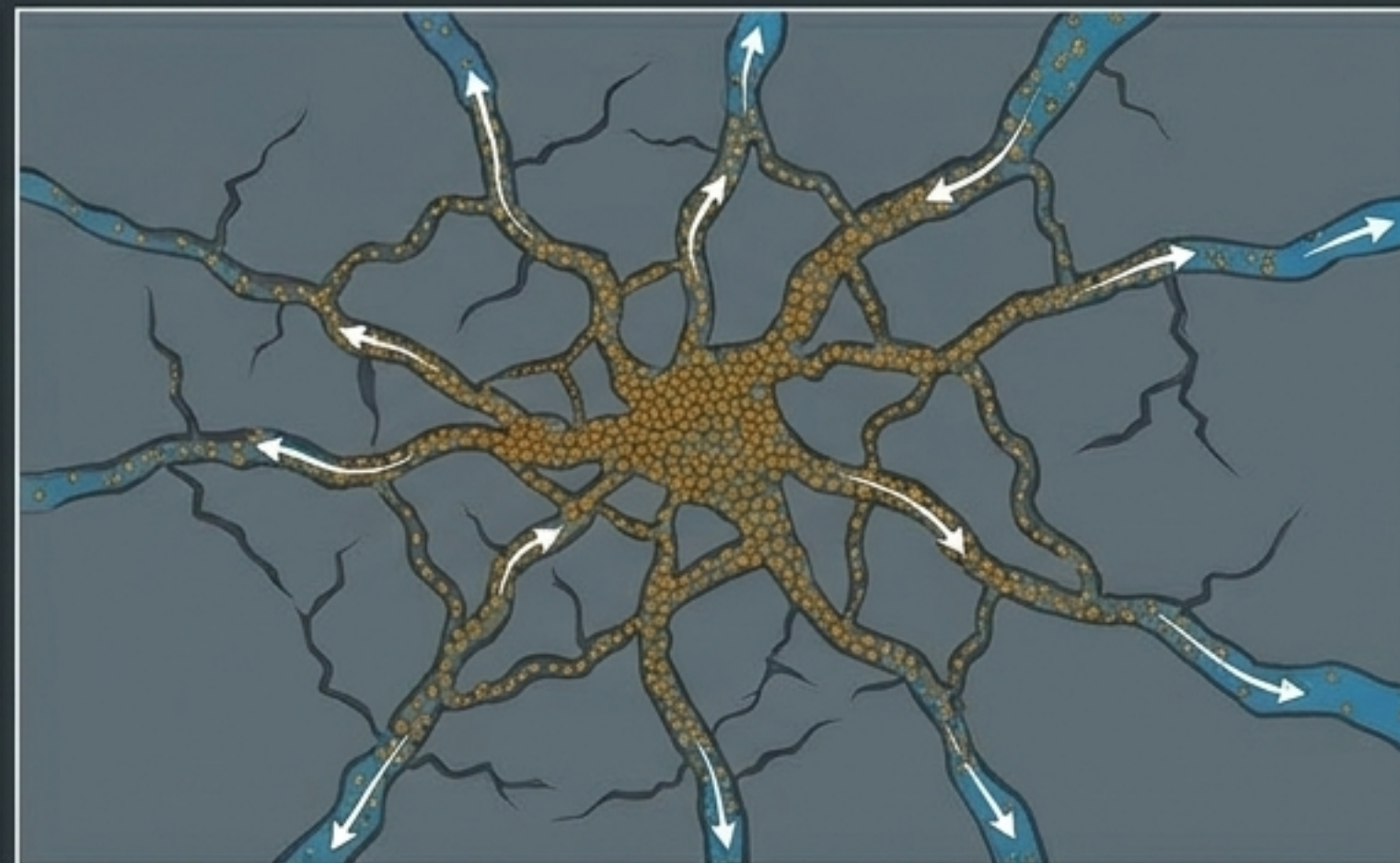
16/30 Mesh (Large Particles)

Designed to hold ultra-wide primary fractures open. Yields maximum permeability and conductivity near the wellbore, but will physically bridge and block narrow pathways.



40/70 or 100 Mesh (Fine Particles)

Designed for extreme reach. Capable of deep penetration into highly complex, extremely tight micro-fracture networks where larger grains would fail.



Synthesis: The Fracture Conductivity Triad

Successful well stimulation is the perfect calibration of three distinct engineering domains. Only when these three elements are perfectly matched to the specific reservoir can sustained hydrocarbon flow be achieved.

